

Rajasthan Heatwave Analysis (2006–2025) Using Python

Abstract

This project analyses weather and heatwave data from Rajasthan between 2006 and 2025. The objective is to study temperature trends, identify heatwave patterns, and understand how weather parameters such as rainfall, cloud cover, and wind affect heatwave occurrences. Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used for data cleaning, analysis, and visualization.

1. Introduction

Heatwaves are prolonged periods of excessively hot weather that can have serious impacts on human health, agriculture, and the environment. Rajasthan is one of the hottest states in India and frequently experiences extreme temperatures during summer.

The purpose of this project is to analyze historical weather data and identify trends related to heatwaves in Rajasthan.

Objectives

- Analyze temperature trends from 2006–2025.
 - Identify the hottest years and months.
 - Examine heatwave frequency.
 - Compare heatwave occurrences across districts.
 - Visualize weather patterns using Python.
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2. Dataset Description

The dataset contains daily weather observations from various districts of Rajasthan.

Features Used

Column	Description
YEAR	Year
MONTH	Month
DAY	Day
TEMP2M	Temperature at 2 meters
TMAX	Maximum Temperature

Column	Description
TMIN	Minimum Temperature
RAIN	Rainfall
CLOUD	Cloud Cover
HEATWAVE	Heatwave Indicator
DISTRICT	District Name

Dataset Size

- Total Records: XXXXX
- Total Columns: 23
- Districts Covered: XX

3. Methodology

Step 1: Data Collection

The dataset was downloaded from Kaggle and imported into Python using Pandas.

Step 2: Data Cleaning

- Checked missing values.
- Removed duplicate records.
- Converted temperature from Kelvin to Celsius.

Step 3: Exploratory Data Analysis

Several visualizations were created to understand temperature and heatwave patterns.

4. Analysis and Results

4.1 Average Temperature Trend

[Insert Chart: Yearly Temperature Trend]

The chart shows how average temperature changed between 2006 and 2025.

Observation

A gradual increase in temperature can be observed in recent years.

4.2 Monthly Temperature Analysis

[Insert Chart: Monthly Average Temperature]

Observation

May and June recorded the highest average temperatures.

4.3 Heatwave Frequency Analysis

[Insert Chart: Heatwave Days Per Year]

Observation

Heatwave events became more frequent in recent years.

4.4 District-wise Heatwave Analysis

[Insert Chart: Top Districts with Heatwaves]

Observation

Certain districts experienced significantly higher heatwave frequencies compared to others.

4.5 Correlation Analysis

[Insert Correlation Heatmap]

Observation

Temperature showed strong relationships with several meteorological variables.

5. Key Findings

1. Average temperatures increased over the study period.
 2. May and June were the hottest months.
 3. Heatwave events were more common after 2018.
 4. Some districts consistently recorded extreme temperatures.
 5. Weather parameters influenced heatwave occurrence.
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6. Conclusion

This project successfully analyzed weather and heatwave trends in Rajasthan from 2006 to 2025. The findings indicate rising temperatures and recurring heatwave events, highlighting the importance of climate monitoring and preparedness.

The project demonstrates the practical application of Python-based data analysis techniques for solving real-world environmental problems.

7. Future Scope

- Heatwave prediction using Machine Learning.
 - Interactive dashboard development using Power BI.
 - Climate comparison with other Indian states.
 - Forecasting future heatwave events.
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References

1. Kaggle Dataset
2. Python Documentation
3. Pandas Documentation
4. Matplotlib Documentation
5. Seaborn Documentation